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ORIGINAL ARTICLE

Box-Behnken modeling to optimize the engineering response and the energy expenditure in material extrusion additive manufacturing of short carbon fiber reinforced polyamide 6

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Abstract

The field of production engineering is constantly attempting to be distinguished for promoting sustainability, energy efficiency, cost-effectiveness, and prudent material consumption. In this study, three control parameters (3D printing settings), namely nozzle temperature, travel speed, and layer height (L_H) are being investigated on polyamide 6/carbon fiber (15 wt%) tensile specimens. The aim is the optimum combination of energy efficiency and mechanical performance of the specimens. For the analysis of the results, the Box-Behnken design-of-experiment was applied along with the analysis of variance. The statistical analysis conducted based on the experimental results, indicated the importance of the L_H control setting, as to affecting the mechanical strength. In particular, the best tensile strength value ($\sigma_B = 83.52$ MPa) came from the 0.1 mm L_H. The same L_H, whereas caused the highest energy consumption in 3D printing ($E_{PC} = 0.252$ MJ) and printing time ($P_T = 2272$ s). The lowest energy consumption ($E_{PC} = 0.036$ MJ) and printing time (PT = 330 s) were found at 0.3 mm L_H. Scanning electron microscopy was employed as a part of the manufactured specimens' 3D printing quality evaluation, while Thermogravimetric analysis was also conducted. The modeling approach led to the formation of equations for the prediction of critical metrics related to energy consumption and the mechanical performance of composite parts built with the MEX 3D printing method. These equations proved their reliability through a confirmation run, which showed that they can safely be applied, within specific boundaries, in real-life applications.

Keywords Polyamide 6 · Carbon fiber · Box-Behnken design · Energy efficiency · Material extrusion (MEX); 3D printing

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Nomenclature

3D-P	Three-dimensional printing
ABS	Acrylonitrile butadiene styrene
AM	Additive manufacturing
ANOVA	Analysis of variance
BBD	Box-Behnken design
CF	Carbon fiber
DOE	Design of experiment
D_P	Degree of freedom for the tested parameter
D_T	Total degree of freedom
E	Tensile modulus of elasticity
E_{PC}	Energy printing consumption
FFD	Full factorial design
F_P	F-value for parameter
HPP	High-performance polymer
L_H	Layer height
MEP	Main effect plot
MEX	Melt extrusion
PA	Polyamide

PA12	Polyamide 12
PA6	Polyamide 6
PC	Polycarbonate
PLA	Polylactic acid
P_T	Printing time
QRM	Quadratic regression model
RQRM	Reduced quadratic regression model
SEM	Scanning electron microscopy
S_{PE}	Specific printing energy
S_{PP}	Specific printing power
SS_E	The sum of squared errors
SS_P	The sum of squared deviations
SS_T	Total sum of squares
TD	Taguchi design
TGA	Thermogravimetric analysis
T_N	Nozzle temperature
T_S	Travel speed
T_t	Tensile toughness
UTS	Ultimate tensile strength
V_e	Variance of error
V_P	Variance of parameter
W_s	Specimen weight
ρ_P	Percentage contribution of each parameter
σ_B	Tensile strength

1 Introduction

Additive manufacturing (AM) is a technology with extended utilization in the industry. It is capable of producing not only complex but also lightweight shapes in a sufficient period of time, while also achieving to minimize the waste of material [1–3], by only using the required amount of it for the construction of the desired part, and its support structure in some cases [4]. AM provides sustainability benefits and aims to produce parts without consuming unnecessary amounts of energy or high machine emissions [3, 5]. Among the fields, AM is applied to automotive [6–9], aerospace [10, 11], as well as biotechnology fields [12–14]. Research in the field often focuses on the effect of the 3D printing parameters on the performance of the parts [15, 16], while modeling tools for the analysis of the experimental data are used [17].

Melt extrusion (MEX) is listed among the most used AM techniques, in polymeric and thermoplastic composite parts manufacturing. It is discerned for its cost efficiency, simplicity, and reduced waste of material [18–21]. MEX has been utilized on a variety of pure [22] or matrix materials in composites [23], but mostly [24] on polylactic acid (PLA) [25–27] and acrylonitrile butadiene styrene (ABS) [28–30] polymers. Among the others, investigations have utilized MEX with polyamides, such as polyamide 6 (PA6) [31–34] or polyamide 12 (PA12) [32, 35, 36] in order to investigate their performance. Polyamides have been also investigated as matrix materials for the development of composites with multi-functional performance [37], even in medical applications [38, 39].

HPPs have been a great influence on the manufacturing of many products [40]. They managed to take the place of metals in fields such as automotive, home appliances, spacecraft, rockets, or even defense systems [41–44] while maintaining a high energy efficiency as well as lightweight products in comparison to the metals [40]. It should be mentioned that HPPs have been favored by the advancement of synthetic chemistry regarding their properties, which became optimized, and their suitability for a wider variety of applications [40]. High-performance polyamides are mostly utilized in the automotive sector, electrical and electronic products, the film market, as well as industrial applications [45].

PA6, as a semicrystalline thermoplastic, is characterized by high strength, great chemical, and wear/abrasion resistance [46]. It is considered an engineering polymeric material suitable for utilization in cases such as automotive parts, electronic devices, or even packaging materials because of the great mechanical and physical properties it possesses [47–49]. It belongs to the engineering plastics possessing the crucial capability of load bearing, self-lubrication, chemical opposition, as well as versatility in various sectors [50]. There are studies based on high-performing polyamide composites for applications with regard to automotive, considering the improved impact failure mechanism they possess, their energy absorbance capability, and their mechanical properties [51]. The mechanical properties, surface quality, and energy efficiency of PA6 fabricated specimens have also been investigated [52]. In another research, the weldability of PA6 plates has been studied under three process parameters and concluded that weld tool pin geometry can affect mechanical strength importantly [53].

The polyamides present low density and have fine strength and heat stability, which leads them to be employed, among other industries, in the aircraft, aerospace, and automotive sectors, next to other HPPs, as mentioned above [54–56]. On the other hand, carbon fiber (CF) can enhance the properties of composites with its beneficial mechanical, thermal, and electrical properties [57]. Polyamide (PA)/CF composites are characterized to be lightweight, with high strength, and modulus of elasticity, wear and corrosion resistance, thermal and electrical conductivity, chemical inertness, and thermal stability [58]. They find applications in lightweight automotive parts and aerostructures [59–61], structural reinforcement of masonry in buildings [62], industrial helmets [63], and civil and structural engineering [64, 65].

Carbon-based materials, such as carbon nanotubes and carbon black, have been investigated for their effect as additives in polyamides [66, 67] or other popular polymers, such as the PLA [68], and ABS [69] in MEX 3D printing, showing a high potential not only as reinforcement agents but also

in inducing multi-functional performance in some cases. 3D printed PA6/CF composites have been investigated as to the effect of microscopic voids on their mechanical performance [31–34]. Information about their crystallization, mechanical, and thermal properties have been obtained and suggested that the addition of CF leads to the increase of tensile modulus and strength along with the decrease of elongations at break [49].

Regression models [70], the Taguchi design of experiment [71], and analysis of variance (ANOVA) [72] are often used as modeling tools for the analysis and optimization of experimental data in 3D printing in studies such as the ones mentioned above. For the experimental analysis of PA and PA-based composites, the Taguchi design [26, 73–75], Box-Behnken design [76–79], and full factorial design [78, 80] have been employed. There is even a study comparing all three experimental designs of them [81]. Box-Behnken design (BBD) versus full factorial design (FFD) has been studied to assess the ultimate tensile strength of PA12 MEX 3D printing specimens, by evaluating three 3D printing settings (input parameters), to be feasible (number of experiments to be conducted) to implement an FFD approach within a study [77].

As AM technology becomes more popular in the manufacturing sector, its sustainability becomes an important issue. The energy for the production of parts with the process is a critical parameter toward the sustainability of the AM technologies. Additionally, sustainability is a factor with increased interest lately worldwide. Herein, the required energy to produce composite parts with the material extrusion (MEX) AM process is assessed and reported. Quantifying the required energy provides information related to various fields. Apart from the sustainability, the cost for the production of a part with the AM method can be more accurately calculated. On the other hand, the mechanical performance of the 3D-printed parts is sensitive to the values of the parameters used with a small change in the parameters' values leading to the production of parts with decreased mechanical properties. This research aimed to provide information on how each parameter affects both the mechanical performance and the energy consumption for the production of composite 3D printed parts. The aim of the research herein is not only to investigate and evaluate the mechanical responses of the PA6/CF fabricated specimens but also to result in developing a combination of suitable process control settings, by minimizing the consumed energy for the fabrication of the parts with MEX 3D printing and maximizing their mechanical performance in the tensile test. The optimum control parameters set should result in a MEX 3D printing procedure characterized by cost-effectiveness, as well as less energy and material consumption while also producing products with high mechanical strength. The BBD design of the experiment analyzed the experimental results. The experimental procedure and the modeling analysis that followed confirmed the hypothesis about the effect and the sensitivity of the 3D printing parameters on both the energy consumption and the mechanical performance of the parts. ANOVA was used to compile prediction models. Their efficiency was confirmed with a confirmation run. The provided prediction models are ready to be used in industrial environments for the calculation of the expected values of critical metrics related to energy consumption and the mechanical performance of composite parts built with the MEX 3D printing method.

2 Materials and methods

A filament of carbon polyamide (1.75 mm) purchased from NEEMA3D™ (the production is located in the Netherlands) was utilized. It is a PA6 material reinforced with 15 wt% carbon fiber. According to the information provided by the supplier, the tensile modulus is 10,500 MPa, the impact strength is 35 kJ/m^2, the mold shrinkage is 0.3–0.5%, and the water absorption is < 0.3%. As for the properties of the filament, its printing temperature ranges between 255 and 265 °C, while its bed temperature ranges between 65 and 75 °C. The filament was fed in an Intamsys Funmat HT apparatus (from Intamsys Technology, located in Shanghai, China). Energy consumption was measured by utilizing a digital multimeter Rigol DM3058E (from RIGOL Technologies, in Beijing, China), while 3D printing time was recorded with the assistance of the stopwatch method [82]. Figure 1 shows the different stages of this study, namely consumption of energy monitoring, the process of 3D printing, evaluation of the samples' morphological characteristics through SEM, tensile testing, and the formulation of the BBD under the three chosen control parameters.

Figure 2b presents the chosen control settings set for the process of 3D printing, for the fabrication of type V tensile specimens, based on the ASTM D638 Standard. There were five samples fabricated for every combination of control parameters (which makes a total of 15 runs, five replicates each). The tensile testing was performed on an Imanda-MX2 tester (from Imanda Inc., in IL, USA) following the ASTM D638-02a standard. The procedure followed was applying a uniaxial force load on the samples for as long as it took for each individual specimen to fail the tensile testing. Stereoscope microscopy was employed for the morphological evaluation of the specimens, by utilizing an OZR5 stereoscope with an ODC 832 digital camera of 5MP (from KERN & SOHN GmbH, Germany) and an SEM (from JSM 6362LV, Jeol, Japan). For the capturing of SEM pictures, the specimens underwent covering with gold-coated (Au) and were into a high vacuum state of 20-kV acceleration voltage during observation. TGA analysis was conducted (Fig. 2a)

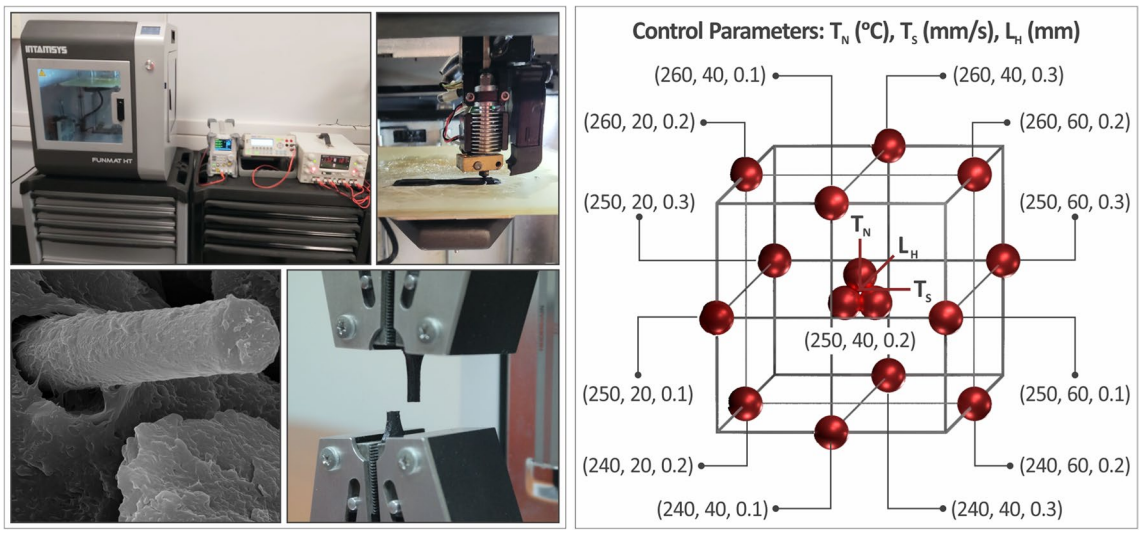

Fig. 1 The various stages of this investigation: (on the left) monitoring of energy consumption, procedure of 3D printing, evaluation of morphological characteristics, tensile testing, and (on the right) brief presentation of Box-Behnken design

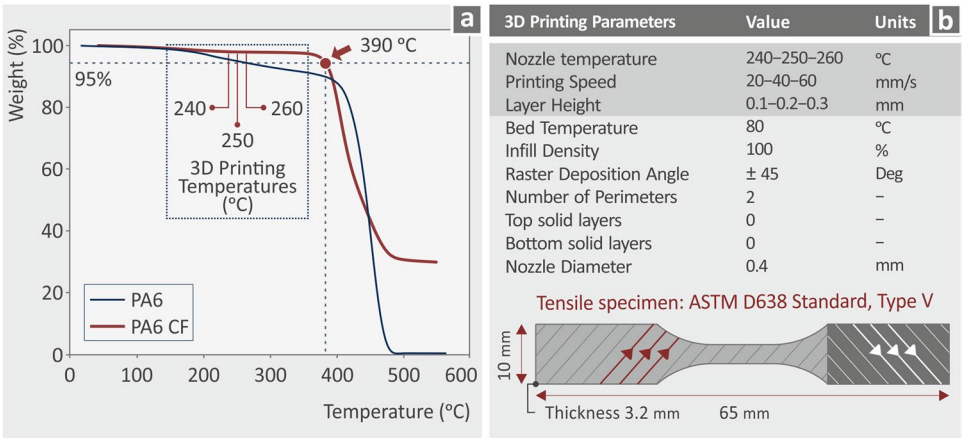

Fig. 2 **a** TGA plot (weight loss % versus temperature) of PA6/CF 15 wt% and pure PA6 given by the producer. **b** Input parameters set for the 3D-P of PA6/CF 15 wt% specimens and geometry of the specimens based on ASTM D638 tensile test standard

with the aim of detecting possible degradation or thermal instability while the 3D printing process of PA6/ CF was run by a TGA Perkin Elmer Diamond TG/TDA apparatus (from PerkinElmer, Inc., Waltham, MA, USA). The cycle of heating ranged between 32 and 390 °C, and the step of heating was 10 °C/min under an atmosphere of nitrogen. The TGA graph of pure PA6, given by the producer, is also presented in Fig. 2a, with the aim of comparing it with PA6/CF 15 wt%. It can be observed that PA6/CF degrades at higher temperatures compared to pure PA6.

2.1 Energy indexes

Three main components were utilized for the calculation of the total consumption and electric energy during the procedure of material extrusion AM. Those were the consumptions deriving from the startup as well as the shutdown of the machine and the consumption deriving from the productive 3D-P process. The following Eq. (1) was utilized for the estimation of the total energy consumption:

$$E_{total} = E_{\text{thermal}} + E_{\text{motion}} + E_{\text{auxiliary}} \tag{1}$$

where E_{thermal} energy consumption is calculated using the following Eq. (2)

$$E_{\text{thermal}} = E_{\text{heating}} + E_{\text{cooling}} \tag{2}$$

E_{motion} represents the energy levels arising from the function of the 3D printers' motors, and $E_{\text{auxiliary}}$ represents the energy consumption arising from the function of electronics as well as the rest fragments of the 3D printer, calculated by the equation below (3):

$$E_{\text{auxiliary}} = E_{\text{startup}} + E_{\text{steadystate}} + E_{\text{shutdown}} \tag{3}$$

Generalizing the energy documentation, S_{PE} (specific printing energy, required energy per mass produced) can be estimated through the equation below (4):

$$S_{PE} = \frac{E_{PC}}{W_S}\left[MJ/g\right] \tag{4}$$

while S_{PP} (specific printing power, required power per mass produced) can be estimated through Eq. (5):

$$S_{PP} = \frac{E_{PC}}{P_T \times W_S} \times 10^6\left[W/g\right] \tag{5}$$

E_{PC} stands for the energy printing consumption which corresponds to the 3D printer (E_{total}), W_S represents the actual value of weight characterizing every specimen, while P_T stands for the printing time consumed so that every experimental run can be conducted.

2.2 DOE, ANOVA, and statistical analysis

Herein, BBD was utilized with the aim of forming a set of optimum input 3D-P settings as to σ_B and E_{PC} of the manufactured material extrusion 3D-P PA6/ CF specimens. The chosen input parameters were nozzle temperature, travel speed, and layer height. These were selected as they were considered the most important ones, based on the literature review carried out on energy consumption vs. mechanical performance of MEX 3D printed parts, presented above. Along with the analysis of the experimental results, main effect plots as well as interaction plots were created, contributing to finding the optimal set of settings. Regression analysis and prediction models were also formed and utilized for the creation of functions regarding the energy indicators of W_S (part weight), P_T (printing time), E_{PC} (energy printing consumption), S_{PE} (specific printing energy), S_{PP} (specific printing power), and the mechanical performance indicators, i.e., σ_B (tensile strength), E (tensile modulus of elasticity), and T_t (tensile toughness). Additionally, two confirmation runs were performed with the aim of validating the prediction quadratic equations.

The ANOVA statistical technique is widely used for the analysis and interpretation of the experimental results while taking into consideration the ratio of each parameter. ANOVA highlights the importance of each parameter in solving the problem. The steps followed in ANOVA for the required calculations are cited below:

SS_T (total sum of squares) was found through Ref [83]:

$$SS_T = \sum_{i=1}^{N}\left(Y_i - \overline{Y}\right)^2 \tag{6}$$

where N represents the total amount of cases of the orthogonal array, and Y_i shows the experimental/ numerical result for the i_{th} experiment,

$$\overline{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i \tag{7}$$

SS_T (total sum of the squared deviations) is calculated by summarizing SS_e (sum of squared error) and SS_P (sum of squared deviations due to every process parameter), so SS_P was defined as [83]:

$$SS_P = \sum_{j=1}^{t}\frac{\left(SY_j\right)^2}{t} - \frac{1}{N}\left[\sum_{i=1}^{N} Y_i\right]^2 \tag{8}$$

where P is one of the parameters, j represents the level number of P, t represents the repetition of each level of P, and SY_j is the sum of the experimental results involving both P and j. SS_e (sum of squares from the error parameters) is [83]:

$$SS_e = SS_T - SS_A - SS_B - SS_C - SS_D - SS_E \tag{9}$$

The total degree of freedom equals $D_T = N - 1$, and the degree of freedom for every tested parameter equals $D_P = N - 1$. The variance of the parameter tested is $V_P = SS_P/D_P$. The *F*-value for every design parameter is simply the ratio of the mean of squares deviations to the mean of the squared error $F_P = V_P/V_e$. The percentage contribution ρ was calculated as [83]:

$$\rho_P = \frac{SS_P}{SS_T} \tag{10}$$

3 Results

3.1 Morphological characterization and analysis of failure

Figure 3 presents stereoscopic images of the fractured surface belonging to one randomly chosen, out of each run's five replicas, specimens of each run. It should be noted that runs 13, 14, and 15 have been repeated under the same conditions, as the Box-Behnken design proposes. It is also observable that there is a brittle structure in all of the samples, which indicates that the specimens were fractured without a remarkable deformation.

The runs selected for extended analysis were runs 1, 9, and 11. Out of the five replicas, one random specimen was chosen. Then their tensile strength to strain graphs were created and are herein presented in Fig. 4a, b, and c, along with their corresponding printing parameters. It can be observed that the value of the T_S printing parameter was chosen to be the same for all runs, while T_N and L_H values ranged, in order to have a better perspective as to the differences of the studied cases. In Fig. 4a, b, and c, the tensile toughness as well as tensile strength values of each run correspondingly are included. There are also microscopic images of

the specimens' side surfaces, respectively, where it can be observed that run 11 (Fig. 4c) presents a thicker sample in comparison to the other two runs. It should be highlighted that the tensile strength of run 11 (Fig. 4c) is explicitly much lower than the other two runs' (Fig. 4b and c). Figure 4d, e, and f show SEM pictures of the fracture surfaces belonging to the corresponding samples of Fig. 4a, b, and c at a 27× magnification. Run 9 (Fig. 4e) seems to present less porosity than runs 1 (Fig. 4d) and 11 (Fig. 4f).

The same samples mentioned above are also shown in SEM images taken from their side surface at 150× magnification (Fig. 5a, c, and e) and their fracture surface at 1000× magnification (Fig. 5b, d, and f). The side surfaces present many voids, and the samples from runs 1 and 11 have a slightly more visible interlayer fusion than the sample from run 9. Porosity and voids are indeed very intense in all of the fracture images, especially in the case of runs 1 and 11. It should be noted that although the only different 3D printing parameter between runs 9 and 11 is the layer thickness, both the side and the fracture images show very different structures, and for instance, run 11 presents a smoother structure than run 9. The presence of the fibers is evident in the fracture surface images. The defects on the side surface indicate that the 3D printing parameters of the specific run were not optimum for the composite. Such voids and defects are expected to negatively affect the mechanical performance of these samples.

3.2 BBD presentation

Table 1 summarizes all input values regarding the three control settings corresponding to each of the fifteen runs, as well as their levels. The mean average values and standard deviations of W_S, P_T, E_{PC}, and σ_B for each run are also shown in Table 1, while the respective values for E, T_t, S_{PE}, and S_{PP} response parameters are presented in Table 2. Additional appendix results can be found in Tables S5, S6, S7, and S8 of the supplementary material.

3.3 Statistical analysis

There were four response parameters for which box plots were created, taking into consideration the aforementioned experimental results, namely printing time (P_T) (Fig. 6a), specimen weight (W_S) (Fig. 6b), tensile strength (σ_B) (Fig. 6c), and energy printing consumption (E_{PC}) (Fig. 6d). The box plots aim to highlight on which response parameter is found the most remarkable influence by each printing parameter.

In the case of P_T, values are found near three values for all control parameters, while on the contrary W_S values appear to be scattered considering T_N, T_S, and L_H. As for σ_B, regarding T_N and T_S, the values are also scattered, but L_H appears to have a great impact on it, as the values of σ_B change importantly along with the L_H fluctuations. E_{PC} values are again gathered mostly around three or two values regarding T_N, T_S, and L_H seem to have a reasonable impact on E_{PC}. Main effect plots (MEPs) are followed ahead and discussed further.

Figure 7 presents MEPs of P_T and W_S (Fig. 7a), σ_B and E_{PC} (Fig. 7b), E and T_t (Fig. 7c), and S_{PE} and S_{PP} (Fig. 7d) as to T_N, T_S, and L_H. Layer height appears to have a great impact on the majority of the response parameters. It causes

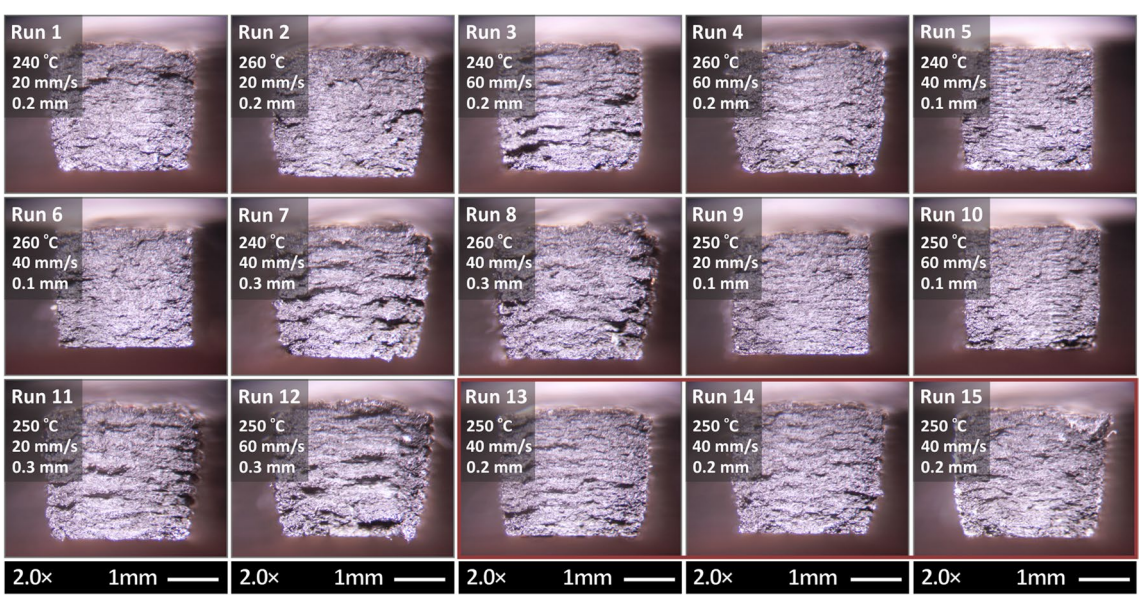

Fig. 3 Stereoscopical images captured from the fractured surface of one random sample chosen from each of the fifteen runs, along with their corresponding 3D-P control settings

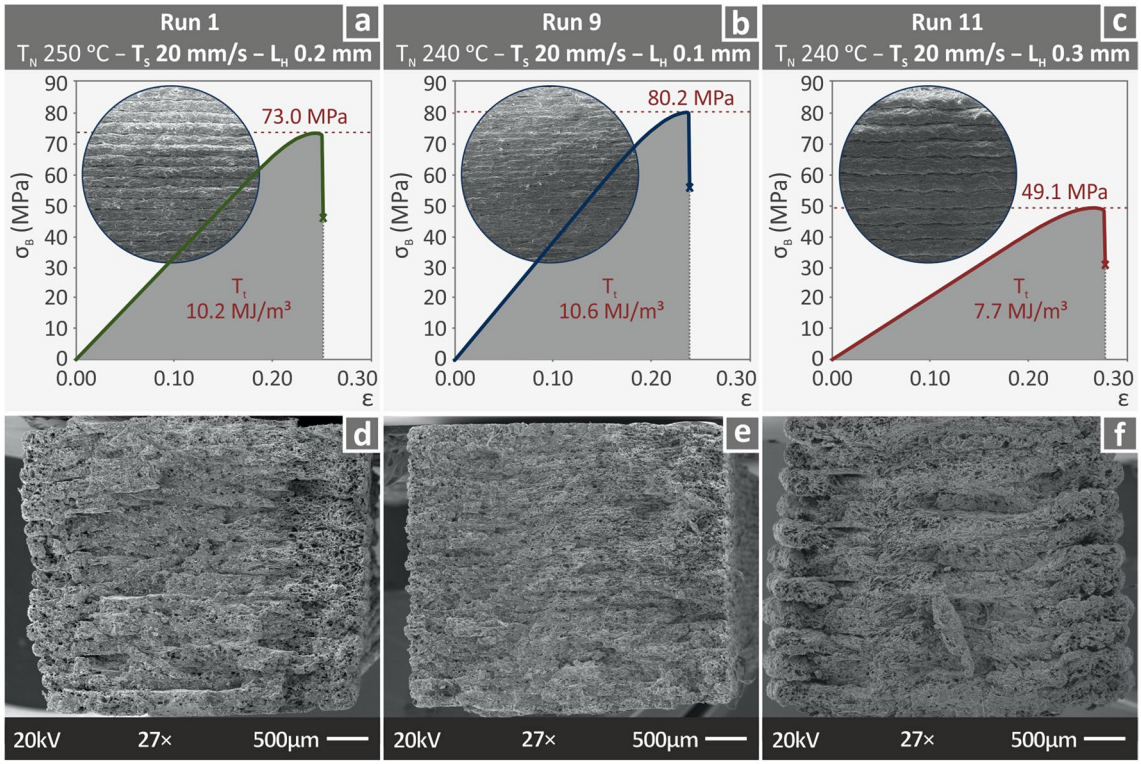

Fig. 4 Side surface SEM images, tensile strength to strain graphs, and the corresponding printing parameters of one random sample selected from run 1 (**a**), run 9 (**b**), and run 11 (**c**), along with SEM images of their fracture surface at 27× magnification: run 1 (**d**), run 9 (**e**), and run 11 (**f**)

an increase in the weight of the specimen, while the printing time is drastically decreased. It also reduces the tensile strength while, at the same time, the printing energy consumption is lowered. This also happens to the tensile modulus of elasticity and tensile toughness. Specific printing energy was decreased under the influence of layer height increase.

Some other remarkable observations are that travel speed lowered printing time as well as specific printing energy and increased specific printing power. Overall, layer height was the control parameter affecting the response parameters the most. Interaction plots of all the input parameters as to the response parameters can be found in Fig. S1 of the supplementary material.

3.4 Analysis of regression results and analysis of variance

QRM of each response can be calculated by the following equation:

$$Y_k = a_k + \sum_{i=1}^{n} b_{i,k} x_i + e_k \tag{11}$$

RQRM of every response can be found by the calculation:

$$Y_k = a_k + \sum_{i=1}^{n} b_{i,k} x_i + \sum_{i=1}^{n} c_{i,k} x_i^2 + e_k \tag{12}$$

where k is the response output (i.e., W_S, P_T, E_{PC}, σ_B, E, T_t, S_{PE}, S_{PP}), a is the constant value, b is the coefficients of the linear terms, c is the coefficients of the square terms, e is the error, and x_i is the three ($n = 3$) control parameters (i.e. T_N, T_S, L_H).

Tables 3, 4, 5, and 6 contain regression values as to the response parameters of W_S, P_T, E_{PC}, and σ_B, while the regression tables of the rest of the response parameters can be found in Tables S1, S2, S3, and S4 of the supplementary information. Along with the tables, equations representing the predictive models formed as a function of T_N, T_S, and L_H for every response parameter are cited (Eqs. 13–20).

The attention is attributed to the F, P, and R^2 values at all tables, as F and P values should be more than 4 and less than 0.05, respectively, and values of R^2 show the % of the accuracy of the model which is expected. All regression values, regarding the total amount of the parameters, were above 85% and the majority was higher than 90%, which indicates that the expected accuracy of the prediction models is remarkably high.

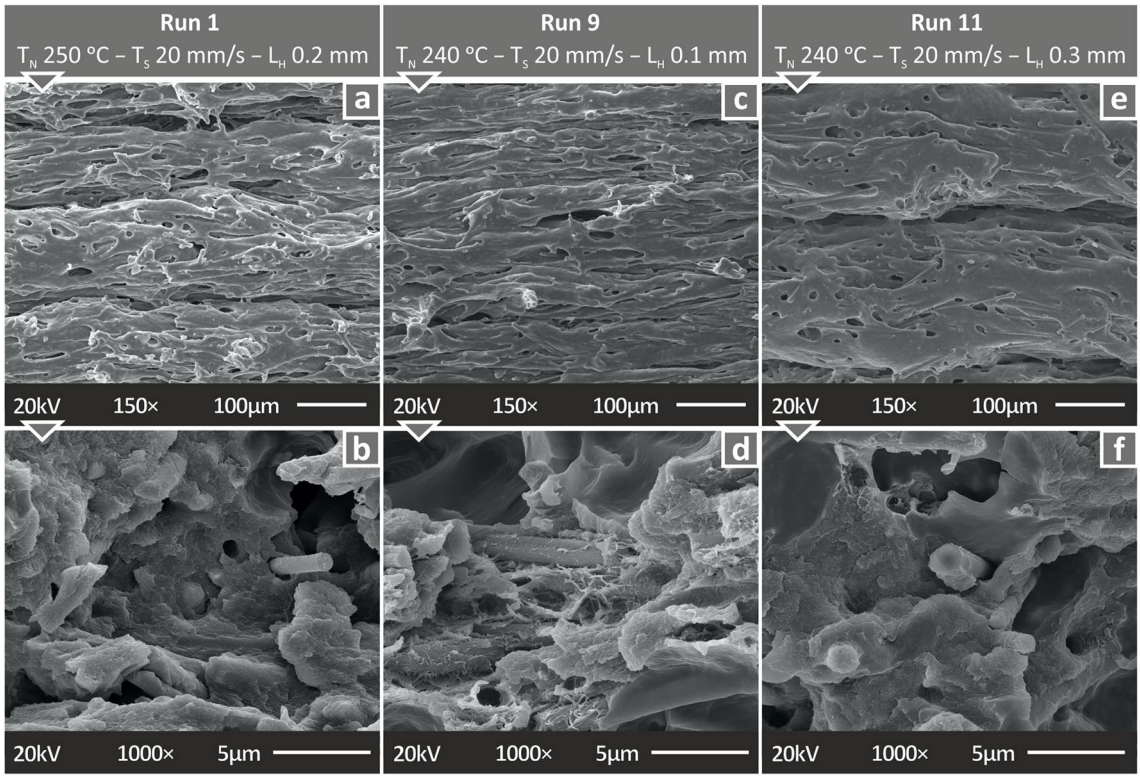

Fig. 5 Side surface SEM images of one random sample selected from run 1 (**a**), run 9 (**c**), and run 11 (**e**) at 150× magnification, along with their corresponding printing parameters and their fracture surface SEM images at a 1000× magnification, run 1 (**b**), run 9 (**d**), and run 11 (**f**)

Table 1 Box-Behnken design: control parameters, levels, mean average values, and standard deviations of measured responses for the W_S, P_T, E_{PC}, and σ_B

Run	T_N	T_S	L_H	W_S (g)	P_T (s)	E_{PC} (MJ)	σ_B (MPa)
1	240	20	0.2	1.99 ± 0.00	1168.60 ± 0.55	0.14 ± 0.00	69.12 ± 3.58
2	260	20	0.2	2.01 ± 0.00	1170.00 ± 0.00	0.14 ± 0.00	65.38 ± 3.59
3	240	60	0.2	1.98 ± 0.00	470.80 ± 0.45	0.07 ± 0.00	60.76 ± 2.69
4	260	60	0.2	2.00 ± 0.00	470.60 ± 0.55	0.07 ± 0.00	52.30 ± 3.35
5	240	40	0.1	1.98 ± 0.00	1220.60 ± 0.55	0.18 ± 0.00	76.91 ± 4.76
6	260	40	0.1	2.00 ± 0.00	1220.60 ± 0.55	0.18 ± 0.00	77.49 ± 4.50
7	240	40	0.3	2.00 ± 0.00	439.60 ± 0.55	0.07 ± 0.00	44.05 ± 1.66
8	260	40	0.3	2.03 ± 0.00	440.00 ± 0.71	0.07 ± 0.00	47.08 ± 2.62
9	250	20	0.1	2.00 ± 0.00	2271.60 ± 0.55	0.25 ± 0.00	76.47 ± 5.26
10	250	60	0.1	1.99 ± 0.01	921.60 ± 0.55	0.14 ± 0.00	76.31 ± 2.83
11	250	20	0.3	2.02 ± 0.01	887.60 ± 0.55	0.11 ± 0.00	47.48 ± 1.35
12	250	60	0.3	2.02 ± 0.01	330.60 ± 0.55	0.06 ± 0.02	44.62 ± 1.68
13	250	40	0.2	2.00 ± 0.00	626.60 ± 0.55	0.07 ± 0.00	56.71 ± 4.89
14	250	40	0.2	2.00 ± 0.00	626.80 ± 0.84	0.07 ± 0.00	58.70 ± 0.89
15	250	40	0.2	2.00 ± 0.00	627.20 ± 0.84	0.07 ± 0.00	55.11 ± 1.08

$$W_S = -1.687 + 0.02858 \times TN - 0.000337 \times TS - 0.2000 \times LH - 0.000055 \times TN^2 + 0.000001 \times TS^2 + 0.8000 \times LH^2 \quad (13)$$

$$P_T = -20,613 + 199 \times TN - 67.23 \times TS - 14,144 \times LH - 0.398 \times TN^2 + 0.5822 \times TS^2 + 24,309 \times LH^2 \quad (14)$$

$$E_{PC} = 7.28 - 0.0540 \times TN - 0.006930 \times TS - 2.2860 \times LH + 0.000108 \times TN^2 + 0.000063 \times TS^2 + 4.320 \times LH^2 \quad (15)$$

Table 2 Mean average values and standard deviations regarding the measured responses of E, T_t, S_{PE}, and S_{PP}

Run	E (MPa)	T_t (MJ/m^3)	S_{PE} (MJ/g)	S_{PP} (W/g)
1	304.50 ± 23.66	9.40 ± 0.89	0.07 ± 0.00	61.86 ± 0.15
2	269.78 ± 17.93	9.51 ± 0.73	0.07 ± 0.00	61.23 ± 0.00
3	280.42 ± 7.62	7.67 ± 0.94	0.04 ± 0.00	77.32 ± 0.17
4	232.55 ± 22.35	7.36 ± 1.61	0.04 ± 0.00	76.50 ± 0.09
5	338.96 ± 31.32	10.21 ± 0.61	0.09 ± 0.00	74.48 ± 0.03
6	382.98 ± 46.63	9.83 ± 1.13	0.09 ± 0.00	73.73 ± 0.03
7	221.92 ± 11.94	5.29 ± 0.64	0.04 ± 0.00	81.81 ± 0.24
8	214.48 ± 11.41	6.57 ± 0.91	0.04 ± 0.00	80.69 ± 0.22
9	374.25 ± 25.85	9.99 ± 1.46	0.13 ± 0.00	55.47 ± 0.01
10	360.49 ± 23.34	9.53 ± 0.85	0.07 ± 0.00	78.36 ± 0.21
11	214.42 ± 8.38	6.68 ± 0.80	0.05 ± 0.00	60.12 ± 0.16
12	215.97 ± 6.33	5.58 ± 0.68	0.03 ± 0.01	86.41 ± 29.53
13	274.06 ± 24.48	7.15 ± 0.89	0.04 ± 0.00	57.45 ± 0.05
14	279.20 ± 9.82	7.88 ± 0.77	0.04 ± 0.00	57.43 ± 0.08
15	260.58 ± 15.72	7.40 ± 0.64	0.04 ± 0.00	57.40 ± 0.08

$$\sigma_B = 124.5 - 0.1072 \times TN - 0.1529 \times TS - 154.93 \times LH \quad (16)$$

$$E = 591 - 0.575 \times TN - 0.459 \times TS - 737.4 \times LH \quad (17)$$

$$T_t = 11.06 + 0.0087 \times TN - 0.03399 \times TS - 19.31 \times LH \quad (18)$$

$$S_{PE} = 3.755 - 0.02788 \times TN - 0.003441 \times TS - 1.1431 \times LH + 0.000056 \times TN^2 + 0.000031 \times TS^2 + 2.152 \times LH^2 \quad (19)$$

$$S_{PP} = 6151 - 48.51 \times TN + 0.079 \times TS - 388.5 \times LH + 0.0969 \times TN^2 + 0.00526 \times TS^2 + 1056 \times LH^2 \quad (20)$$

There were some more graphs created in conjunction with the regression data, in order to highlight the important parameters of this study. Figures 8 and 9 contain two types of graphs; one graph indicates statistically which parameters can influence the response parameters, and the other graph presents the experimental versus the predicted values regarding the response parameters. Figure 8a includes the corresponding graphs of P_T, where on the left it is shown that L_H and T_S are statistically important, and on the right, it is shown that the experimental and predicted values are importantly close or almost the same (including the confirmation runs). Figure 8b has the graphs of W_S, where on the left, it is shown that all three control parameters T_N, L_H, and T_S are statistically important, and on the right, it is shown that some of the experimental and predicted values are close or almost the same, but some others diverge.

In Fig. 9a the corresponding graphs of σ_B are presented. On the left, it is shown that L_H and T_S are statistically important, while on the right, it is shown that some of the experimental and predicted values (including those of the experimental runs') are importantly close or almost the same. Figure 9b presents the graphs of E_{PC}, where on the

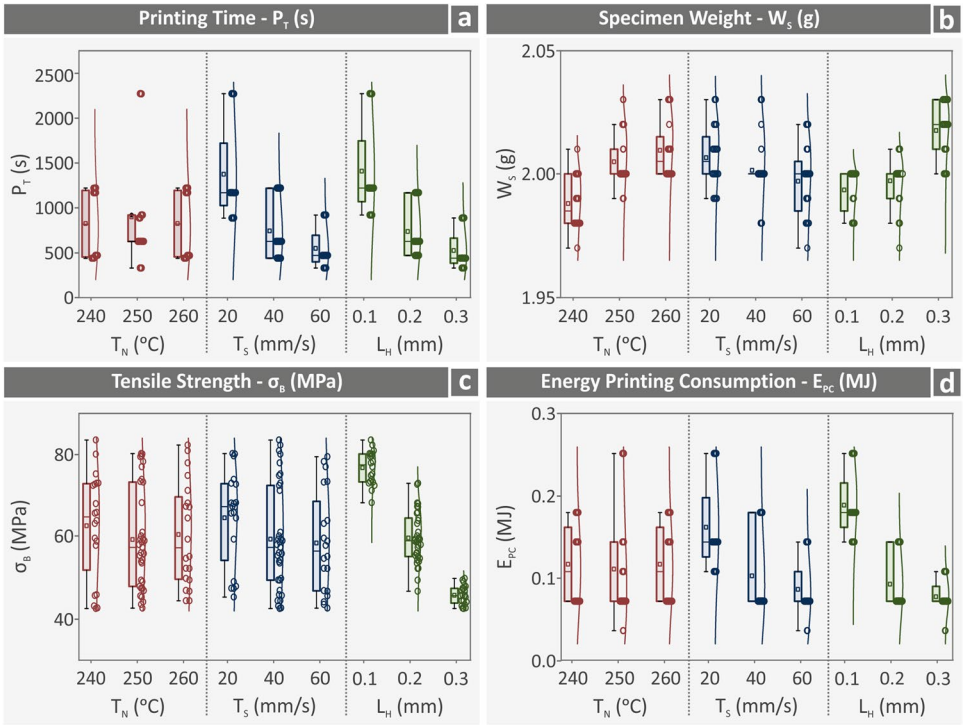

Fig. 6 Box plots presenting the metrics of the main response parameters as to the control settings (T_N, T_S, L_H): P_T (**a**), W_S (**b**), σ_B (**c**), and E_{PC} (**d**)

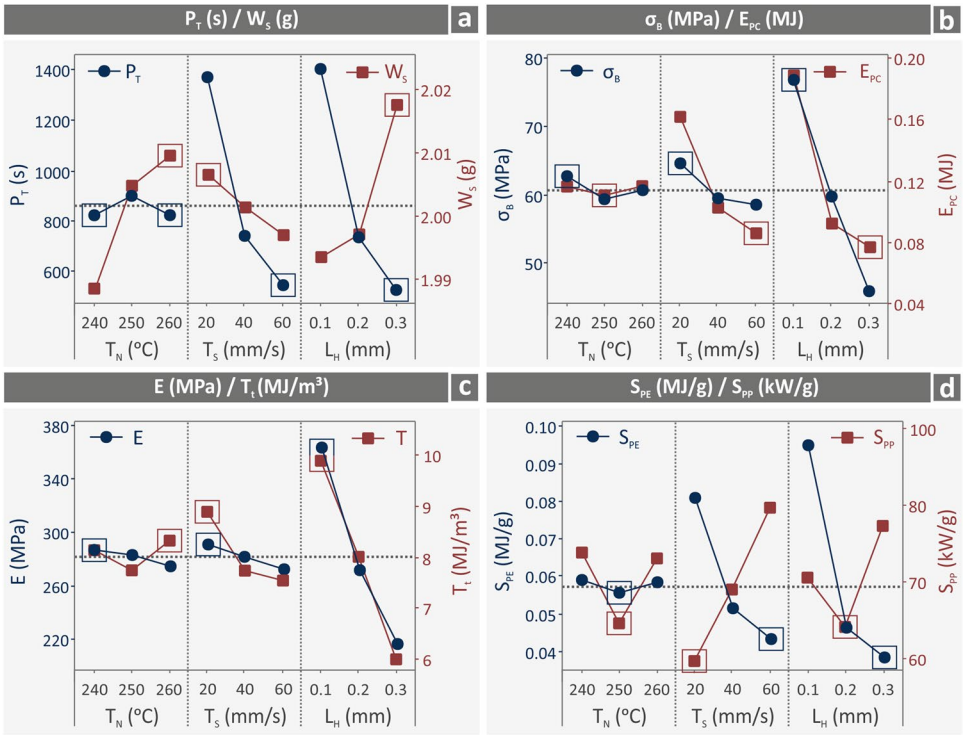

Fig. 7 Main effect plots from all of the response parameters as to the control parameters (T_N, T_S, L_H): P_T and W_S (**a**), σ_B and E_{PC} (**b**), E and T_t (**c**), S_{PE} and S_{PP}(**d**)

Table 3 Polynomial ANOVA, specimen W_S versus T_N, T_S, and L_H

Source	DF	Adj SS	Adj MS	*F*-value	*P*-value
Regression	6	0.013163	0.002194	176.55	0.000
T_N	1	0.000603	0.000603	48.51	0.000
T_S	1	0.000051	0.000051	4.11	0.046
L_H	1	0.000449	0.000449	36.10	0.000
T_N^2	1	0.000558	0.000558	44.94	0.000
T_S^2	1	0.000005	0.000005	0.37	0.544
L_H^2	1	0.001182	0.001182	95.08	0.000
Error	68	0.000845	0.000012		
Total	74	0.014008	0.002206		
$\boldsymbol{R}^2$	93.97%				
$\boldsymbol{R}^2$(adj)	93.44%				
$\boldsymbol{R}^2$(pred)	92.55%				

Table 4 Polynomial ANOVA, P_T versus T_N, T_S, and L_H

Source	DF	Adj SS	Adj MS	*F*-value	*P*-value
Regression	6	16,685,333	2,780,889	179.19	0.000
T_N	1	29,183	29,183	1.88	0.175
T_S	1	2,027,581	2,027,581	130.65	0.000
L_H	1	2,243,714	2,243,714	144.57	0.000
T_N^2	1	29,183	29,183	1.88	0.175
T_S^2	1	1,001,327	1,001,327	64.52	0.000
L_H^2	1	1,090,958	1,090,958	70.30	0.000
Error	68	1,055,334	15,520		
Total	74	17,740,667	2,796,409		
$\boldsymbol{R}^2$	94.05%				
$\boldsymbol{R}^2$(adj)	93.53%				
$\boldsymbol{R}^2$(pred)	92.66%				

left, it is shown that L_H and T_S are statistically important, and on the right, it is shown that some of the experimental and predicted values mostly coincide (including the confirmation runs). In Fig. 9a and b, two SEM images of the confirmation run, one from the fracture surface (Fig. 9a) and one from the side surface (Fig. 9b), are included. Additional results regarding the corresponding graphs of E, T_t, S_{PE}, and S_{PP} can be found in Fig. S2 and Fig. S3 of the supplementary information.

Figure 10 presents 3D surface plots of the σ_B and E_{PC} response parameters as to the T_N, T_S, and L_H control parameters. Figure 10a, b, and c show the graphs of σ_B as to T_N and T_S, T_N and L_H, and T_S and L_H correspondingly. It can be observed that lower T_N and T_S impart higher tensile strength, while the opposite happens in the case of higher T_N and T_S. The effect of L_H seems to be strong, as it is obvious that its decrease causes a rise of σ_B, and the opposite, at both high and low T_N or T_S. Figure 10d, e, and f depict the graphs of E_{PC} as to T_N and T_S, T_N and L_H, and T_S and L_H, respectively. Low T_S and L_H seem to cause the increase of E_{PC} and the opposite, at both high and low T_N. In addition, a combination of low L_H and T_S increases the E_{PC} and the opposite. More data regarding surface plots of P_T and W_S response parameters can be found in Fig. S4 of the supplementary information.

Table 5 Polynomial ANOVA, E_{PC} versus T_N, T_S, and L_H

Source	DF	Adj SS	Adj MS	*F*-value	*P*-value
Regression	6	0.225694	0.037616	429.06	0
T_N	1	0.002153	0.002153	24.56	0
T_S	1	0.021544	0.021544	245.74	0
L_H	1	0.058607	0.058607	668.49	0
T_N^2	1	0.002153	0.002153	24.56	0
T_S^2	1	0.011724	0.011724	133.73	0
L_H^2	1	0.034454	0.034454	392.99	0
Error	68	0.005962	0.000088		
Total	74	0.231656	0.037704		
R^2	97.43%				
R^2(adj)	97.20%				
R^2(pred)	96.82%				

Table 6 Polynomial ANOVA, σ_B versus T_N, T_S, and L_H

Source	DF	Adj SS	Adj MS	*F*-value	*P*-value
Regression	3	10021.1	3340.35	173.65	0.000
T_N	1	46.0	46.00	2.39	0.126
T_S	1	374.0	373.97	19.44	0.000
L_H	1	9601.1	9601.08	499.11	0.000
Error	71	1365.8	19.24		
Total	74	11386.9	3359.59		
R^2	88.01%				
R^2(adj)	87.50%				
R^2(pred)	86.67%				

3.5 Confirmation experiments

Table 7 summarizes the input values regarding the three control settings corresponding to the two confirmation runs, as well as their levels. The mean average values and standard deviations for the confirmation runs of W_S, P_T, E_{PC}, and σ_B for each run are also shown in Table 7, while the respective values for E, T_t, S_{PE}, and S_{PP} response parameters are presented in Table 8. Table 9 presents the validity results, comparing the predicted with the actual values of the response parameters in the confirmation runs. Overall, the % deviation between the predicted with the actual values is exceptionally low, showing that the prediction models in this study are expected to provide very accurate and reliable results. The only exception is the deviation of the E and especially the S_{PE} values of run 17. Such deviations indicate that the prediction models operate within a specific range of values and probably the control values of run 17 are marginal for the capacity of the prediction models of these two response metrics. The analytic results of the experimental course of the confirmation runs are presented in the supplementary information provided.

4 Discussion

This study aimed to achieve both increased mechanical properties and minimized energy printing consumption when 3D printing PA6/CF 15 wt% parts with the MEX process. There were three (3) generic 3D printing control parameters

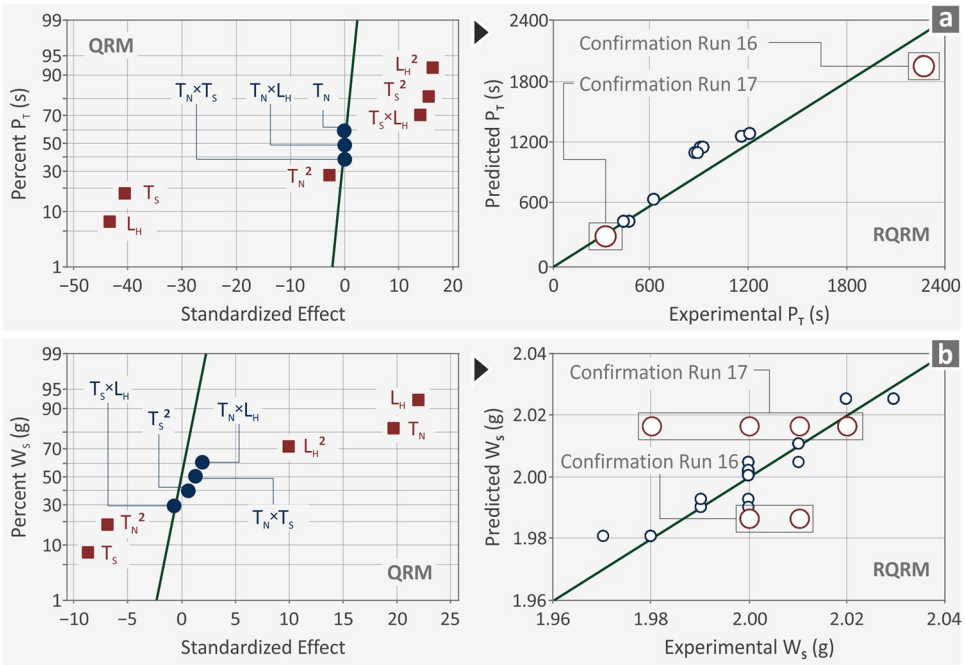

Fig. 8 Graphs of statistical presentation for the % of influence control parameters have on response parameters (on the left) and predicted versus experimental results (on the right), for P_T (**a**) and for W_S(**b**)

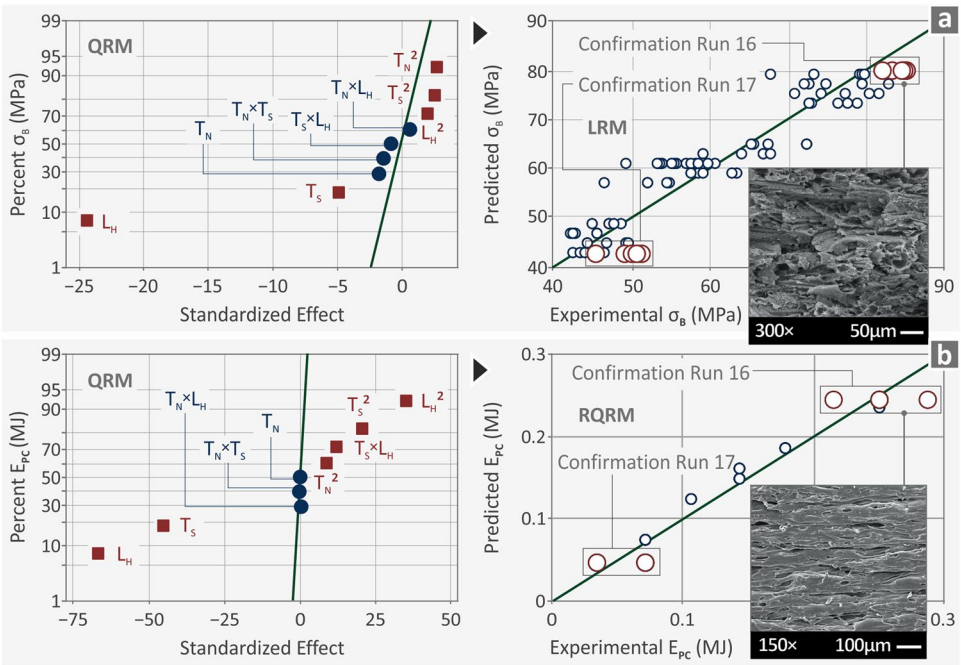

Fig. 9 Graphs of statistical presentation for the % of influence control parameters have on response parameters (on the left) and predicted versus experimental results (on the right), for σ_B (**a**) and for E_{PC}(**b**), along with SEM images from the confirmation runs' structure and side surface correspondingly

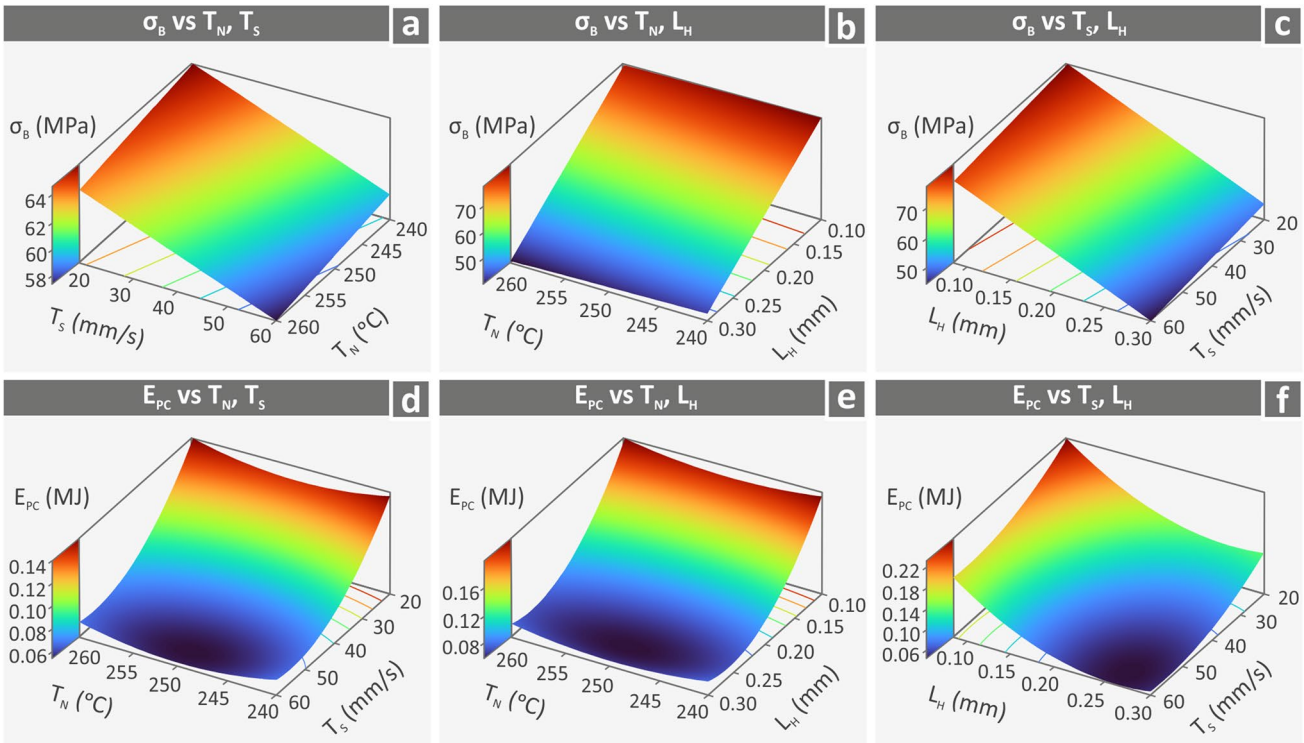

Fig. 10 Surface graphs of σ_B versus T_N and T_S (**a**), T_N and L_H (**b**), T_S and L_H (**c**), and E_{PC} versus T_N and T_S (**d**), T_N and L_H (**e**), T_S and L_H (**f**)

investigated, namely T_N, T_S, and L_H, and eight (8) response parameters, namely W_S, P_T, E_{PC}, σ_B, E, T_t, S_{PE}, and S_{PP}. The 3D-P specimens went through an experimental procedure, and their results underwent analysis and evaluation according to the Box-Behnken design of experiments.

The analysis suggested that the parameter having the most important influence on the mechanical behavior of the 3D-P specimens was L_H, which, to the authors' best knowledge, has not been investigated yet in the literature for the specific composites in the MEX process. Tensile strength

Table 7 Control parameters, levels, mean average values, and standard deviations of measured responses of W_S, P_T, E_{PC}, and σ_B for the confirmation runs

Run	T_N	T_S	L_H	W_S (g)	P_T (s)	E_{PC} (MJ)	σ_B (MPa)
16	240	20	0.1	2.00 ± 0.01	2271.80 ± 0.84	0.25 ± 0.03	84.52 ± 1.28
17	250	60	0.3	2.01 ± 0.02	331.00 ± 0.71	0.06 ± 0.02	49.45 ± 2.37

Table 8 Mean average values and standard deviations of measured responses of E, T_t, S_{PE}, and S_{PP} for the confirmation runs

Run	E (MPa)	T_t (MJ/m^3)	S_{PE} (MJ/g)	S_{PP} (W/g)
16	463.19 ± 5.76	10.44 ± 0.63	0.13 ± 0.01	55.36 ± 5.67
17	272.92 ± 9.90	5.39 ± 0.62	0.03 ± 0.01	86.94 ± 30.30

Table 9 Validation results

	Run:	16	17
Actual	W_S (g)	2.00	2.01
	P_T (s)	2271.80	331.00
	E_{PC} (MJ)	0.25	0.06
	S_{PE} (MJ/g)	0.13	0.03
	S_{PP} (W/g)	55.36	86.94
	σ_B (MPa)	84.52	49.45
	E (MPa)	463.19	272.92
	T_t (MJ/m^3)	10.44	5.39
Predicted	W_S (g)	1.99	2.02
	P_T (s)	1939.17	268.73
	E_{PC} (MJ)	0.24	0.04
	S_{PE} (MJ/g)	0.14	0.04
	S_{PP} (W/g)	65.43	81.92
	σ_B (MPa)	80.22	42.05
	E (MPa)	370.08	198.49
	T_t (MJ/m^3)	10.54	5.40
Absolute error	W_S (%)	0.91	0.49
	P_T (%)	14.64	18.81
	E_{PC} (%)	3.97	23.61
	S_{PE} (%)	11.47	42.15
	S_{PP} (%)	18.20	5.78
	σ_B (%)	5.08	14.97
	E (%)	20.10	27.27
	T_t (%)	0.92	0.19

and 3D printing energy consumption were only some of the response parameters considerably affected by L_H. Additionally, 3D printing time and specimens' weight were also much affected, as well as tensile toughness and tensile modulus of elasticity. The tensile strength ranged between 42.59 and 83.52 MPa (about 100% difference), while the 3D printing energy consumption managed to reach the lowest value of 0.036 MJ (with the highest one being also 0.19 MJ, 5.2 times more). Such differences show the importance of selecting proper 3D printing settings, as they could have a quite large effect on the performance of the process and the parts produced. Such differences also justify the need for such an analysis as the one conducted in the study, as it was found that both the sustainability of the process and the mechanical performance of the parts could be highly affected.

As the L_H increased, and so did the weight of specimens, the required 3D printing time was found to be reduced, along with the required 3D printing energy and specific printing energy, which reached the lowest values of 330 s, 0.036 MJ, and 0.018 MJ/ g, respectively, at the highest L_H of 0.3 mm acting beneficially. However, the same value of L_H resulted in the lowest tensile strength of 42.59 MPa, tensile modulus of elasticity of 197.77 MPa, and tensile toughness of 4.57 MJ/ m^3.

On the other hand, the lowest value set for the L_H equal to 0.1 mm had the exact opposite influence on all of the aforementioned response parameters. Although the 3D printing time was increased at 2272 s, the 3D printing energy at 0.252 MJ, and specific printing energy at 0.126 MJ/ g, the lower L_H acted beneficial for the tensile strength which reached 83.52 MPa, tensile modulus of elasticity 427.42 MPa, and tensile toughness 11.96 MJ/ m^3. It should be mentioned that the weight of specimens was also reduced.

The rest of the control settings (T_N and T_S) did not have such a remarkable impact on the response parameters as the L_T had. T_N caused an increase in the weight of the specimen, as it got higher, while the rest of the response parameters were not much affected. On the other hand, T_S influenced a few more response parameters, such as the 3D printing time, which was reduced the higher T_S was, as well as the tensile strength and specific printing energy, which were slightly reduced by its increase. Additionally, the specific 3D printing power became higher with the rise of T_S. Studies on energy consumption vs. mechanical performance in the 3D printing of parts with the MEX process from different polymeric materials (PLA [27] and ABS [30]) agree with the findings of the current research. They have also reported that the travel speed and the layer height are the 3D printing settings mostly affecting both the energy consumption and the mechanical performance of the 3D printed parts.

There was not a set of parameters that combined both excellent mechanical strength and minimized energy consumption. Studies found in the available literature have

investigated the effect of CF reinforcing PA6 and showed that it provides excellent mechanical properties and can improve both tensile modulus by 60% and tensile strength by about 50% [49]. In [84] investigation, it was proved numerically and experimentally that the rectangle infill pattern with 40% infil density achieved the lowest distortions.

Further analysis was conducted through the examination of SEM images captured from the specimens' fracture and side surfaces at different magnifications. It should be noted that those SEM images have indicated the existence of reinforced short carbon fibers appearing in the form of grayish cylinders (Fig. 5b and f). This has also been reported in other studies in the case of PLA/CF [85], short CF-reinforced PA composites [86, 87], and CF-reinforced PA6 composites [88].

In this study, it was also observed through the combination of SEM images and experimental results that the sample from run 9 (Fig. 5d), which presented the least porosity, had the highest tensile strength. Runs 1 and 11 indicated more porosity, and their tensile strength was found to be lower, which also happened in the case of short CF-reinforced PC polymer matrix composite material in another study [89].

Overall, the constitution of the optimal set of control settings combining both the greatest energy efficiency and mechanical strength was not achieved. Yet an equanimous set of control settings could be derived from this study, as there are parameters that can level off the various influences on each aspect. Moreover, it should be mentioned that the created equations were proved reliable through the calculation of factors with the assistance of ANOVA, as well as the confirmation runs conducted so that they could be considered reliable. This indicates the suitability of the Box-Behnken method for the analysis of the experimental data in this specific study.

5 Conclusions

Herein, the optimal set of control settings is being researched with the challenge of manufacturing parts both energy efficient and mechanically strong, out of PA6/CF 15 wt% composite and through MEX 3D-P. The energy consumption was measured as the specimens were being printed, which then underwent tensile testing coherent with the ASTM D638 standard. Subsequently, the Box-Behnken design was utilized for the examination and analysis of the control settings and how they interact with each other. In addition, there were prediction models created for the total amount of the metrics along with the conduction of two confirmation runs, in order to verify the reliability of the model.

There was not a pair of parameters' values that could achieve both having the lowest energy consumption and the highest mechanical strength. This event compels the need to choose the most desired aspect depending on the current requirements of the application. The only case where the tensile strength presented a medium value and the energy consumption was considerably low, was when a higher travel speed was applied. The most influencing control parameter was the layer height which was proved to affect all of the response parameters. The rest of the settings only had a small effect on the performance of the samples. Further investigation in the future could include the examination of a different set of input and control parameters in order to achieve the optimization of energy efficiency and great mechanical performance of the desired 3D printed parts simultaneously. Additional mechanical tests can be carried out and the range of the control parameter levels can be broadened.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00170-024-13617-5.

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Data availability The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

Declarations

Competing interests The authors declare no competing interests.

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